

$$2) \begin{bmatrix} 1 & -2 & 1 \\ 4 & -7 & 3 \\ -2 & 6 & -4 \end{bmatrix} \quad \text{Solution:} \quad \begin{bmatrix} 1 & -2 & 1 & 1 & 0 & 0 \\ 4 & -7 & 3 & 0 & 1 & 0 \\ -2 & 6 & -4 & 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} \text{Row 2} = \text{Row 2} - 4 \times \text{Row 1} \\ \text{Row 3} = \text{Row 3} + 2 \times \text{Row 1} \end{array}$$

$$\begin{bmatrix} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -4 & 1 & 0 \\ -2 & 6 & -4 & 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} \text{Row 3} = \text{Row 3} + 2 \times \text{Row 2} \\ \text{Row 1} = \text{Row 1} + 2 \times \text{Row 2} \end{array}$$

$$\begin{bmatrix} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -4 & 1 & 0 \\ 0 & 0 & -2 & 10 & -2 & 1 \end{bmatrix} \quad \begin{array}{l} \text{Row 1} = \text{Row 1} + 2 \times \text{Row 2} \\ \text{left of} \end{array}$$

Verification of whether the final matrix is in ~~ref~~ rref form

1. Row 1 and Row 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all the non-zero rows is above the zero rows is True.
 2. Row 1 Leading entry = 1 in column 1
Row 2 Leading entry = 1 in column 2
 $\therefore$ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.
 3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
 $\therefore$ The condition that for each column having a leading entry, all positions below it must contain 0 is True.
 4. The condition that each leading entry must be 1 is True.
 5. Row 1 Column 2 = 0
 $\therefore$ The condition that for each column having a leading entry, all positions above it must contain 0 is True.
- So, the final matrix is in rref. II
- $\therefore$ The original matrix cannot be row-reduced to I. By Theorem 7, the given matrix has no inverse.
- Exercises 27 and 28 prove special cases of the facts about elementary matrices stated in the box following Example 5. Here A is a 6×3 matrix and $I = I_3$. (A general proof would require slightly more notation.)